भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 306: Machine Learning Lab
Practice Assignment 4

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. (a) Write a program to create a synthetic dataset of 100 patterns for car reselling price with three numerical variables as below:
    1. price (y): Target variable (in dollars/ Rs.).
    2. mileage (x_1): Independent variable.
    3. Age of the car (x_2): Independent variable (can be calculated as ((current year) - (year of manufacture)), where current year is 2026.

   (b) Split the dataset into training set: testing set with the ratio as below:
   $training$ = 20:20:80 (i.e., initial: 20%, increment by 20%, maximum is 80%)
   $testing$ = 80:20:20 (i.e., initial: 20%, decrement by 20%, minimum is 20%).

   (c) Train multiple linear regression models for each training: testing cases from 1(b) using Batch Gradient Descent (GD) method, **without using in-built Python packages /libraries** for estimation of the weights/ parameters. Consider the weights (w_0, w_1, w_2)are initialized to zero and the learning rate (α) is set to 0.01. Separately save the model parameters / wights for all the training splits.

   (d) Visualize and store the different plots (best-fit-line) for the different hypotheses designed/ obtained ($\hat{y}$).

   (e) Calculate and store the house price for the testing patterns over different test splits considering the respective hypothesis.

   (f) Calculate the mean squared error (MSE), and coefficient of determination (R^2) for different test splits separately.

   (g) Select the best training-testing split (in terms of MSE). Using the selected split, calculate the model parameters and design hypotheses, considering various w-initializations as zero and random values in the range [0,1], and learning rates (α) as 0.0001, 0.05, 0.1, 1. Store the estimated weights (w_0, w_1, w_2), (R^2) and MSE scores for all the possible combinations of w-values and learning rates.

   (h) Report and interpret the results of 1(c)-1(g) in text file. Store the plots in a folder.

2. Execute the above assignment (in question number 1) in-built package/library for Gradient Descent.